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An assessment of regional vulnerability of rice to climate change in India

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Abstract A simulation analysis was carried out using the InfoCrop-rice model to quantify impacts and adaptation gains, as well as to identify vulnerable regions for irrigated and rain fed rice cultivation in future climates in India. Climates in A1b, A2, B1 and B2 emission scenarios as per a global climate model (MIROC3.2.HI) and a regional climate model (PRECIS) were considered for the study. On an aggregated scale, the mean of all emission scenarios indicate that climate change is likely to reduce irrigated rice yields by ~4 % in 2020 (2010–2039), ~7 % in 2050 (2040–2069), and by ~10 % in 2080 (2070–2099) climate scenarios. On the other hand, rainfed rice yields in India are likely to be reduced by ~6 % in the 2020 scenario, but in the 2050 and 2080 scenarios they are projected to decrease only marginally (<2.5 %). However, spatial variations exist for the magnitude of the impact, with some regions likely to be affected more than others. Adaptation strategies comprising agronomical management can offset negative impacts in the near future—particularly in rainfed conditions—but in the longer run, developing suitable varieties coupled with improved and efficient crop husbandry will become essential. For irrigated rice crop, genotypic and agronomic improvements will become crucial; while for rainfed conditions, improved management and additional fertilizers will be needed. Basically climate change is likely to exhibit three types of impacts on rice crop: i) regions that are adversely affected by climate change can gain in net productivity with adaptation; ii) regions that are adversely affected will still remain vulnerable despite adaptation gains; and iii) rainfed regions (with currently low rainfall) that are likely to gain due to increase in rainfall can further benefit by adaptation. Regions falling in the vulnerable category even after suggested adaptation to climate change will require more intensive, specific and innovative adaptation options. The

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present analysis indicates the possibility of substantial improvement in yields with efficient utilization of inputs and adoption of improved varieties.

1 Introduction

With the threat of climate change looming large on crop productivity, the most vulnerable regions of the world are the tropics. In semi-arid regions higher temperatures and rainfall variability could also significantly affect agricultural production with large implications on food security (IPCC 2007). On the other hand, crop production needs to be doubled by 2050 to meet the demands of an increasing population and economic growth in developing countries (FAO 2009). As a staple food for half of the global population—particularly from Asia and some parts of Africa and Latin America (IRRI 2012)—rice is the most important food grain. Out of 672 Mt of global rice (paddy) production in 2010, Asian countries accounted for about 90 % of the total production; and about half of this came from China and India (FAOSTAT 2012). Rice is the most important staple crop in India; and out of the annual food grain production of 241.2 Mt during 2010–2011, rice (grain) contributed ~39.4 % (95.33 Mt) of the annual production with an area of 42.56 Mha and productivity of 2.24 $Mg ha^{-1}$ (DES 2012). Apart from being consumed as a staple diet, rice also contributes to the livelihood of millions of farmers, the bulk of whom is resource poor. The rural economy is in fact appreciably interlocked with the performance of rice; and improvement in rice yields has significantly helped to reduce rural poverty in India and China—the world's two most populated countries (Fan and Chan 2005).

In India the monsoon (June-October; *Kharif* crop) is the major season for rice cultivation, contributing ~86 % to the annual production, while the remaining 14 % is contributed by the winter (November-March; *Rabi* crop) and summer (March-June; *Zaid* crop) harvests of rice (http://www.rkmp.co.in/rice-almanac). The monsoon's behaviour, therefore, primarily influences crop production in India. For instance, in 2009–10 the late onset of monsoon rainfall and its intra-seasonal variability caused a reduction of ~10 Mt in rice (grain) production. Climate change is thus projected to increase climatic stresses on crops; and a global analysis has projected that an increase in temperatures is likely to affect crop production by 10–40 % in India by 2080–2100 (IPCC 2007) under business as usual scenario.

Climate change influences rice crop mainly through increased atmospheric CO_2, temperature and change in rainfall. Even though the magnitude of CO_2 fertilization benefits on rice plant is debatable, free air CO_2 enrichment (FACE) experiments indicate a gain in yield by ~7–15 % at 570 $\mu mol mol^{-1}$ for the *japonica* type cultivar (Kim et al. 2003a, b). In India, *indica* cultivars are generally grown, which have shown almost similar responses to elevated levels of CO_2 (Moya et al. 1998; Hatfield et al. 2011). The net effect of elevated CO_2 on rice growth and yields are temperature-dependent (Moya et al. 1998; Baker 2005). High temperature-associated yield reduction has been due to increased spikelet sterility (Matsui et al. 2000), decreased sink formation, increased maintenance respiration, and shortened crop duration (Matthews and Wassmann 2003). Previous studies based on data from a few locations with a fixed increase in temperature (1–5 °C) and CO_2 (350–850 $\mu mol m^{-2} s^{-1}$) levels—or with coarse resolution global climate model (GCM) outputs—have projected a decrease in rice yields in India due to climate change (Aggarwal and Mall 2002; Krishnan et al. 2007). However, these point-based simulation impact studies had been extrapolated to large regions, which may have led to huge uncertainties. These studies, moreover, did not attempt to provide possible adaptation strategies either.

Unlike earlier studies, the use of gridded data in this particular study has helped to delineate regional variations more precisely as against the use of point data from a few locations across India. Since estimates indicate that by 2020 food grain demand will increase by 30–50 % over that of 2000 (Paroda and Kumar 2000), the present simulation was carried out with specific objectives to i) quantify impacts of climate change on irrigated and rainfed rice yields at regional level; ii) assess adaptive gains from low-cost and simple strategies; and iii) work-out vulnerable regions for rice production. General adaptation strategies in crop management are improved varieties of crop and agronomy, including efficient use of fertilizers and water (Smit and Skinner 2002). The present study also identified vulnerable areas for need-based technology development. For this purpose climate outputs of a GCM (MIROC3.2.HI, Atmosphere and Ocean Research Institute, Japan; National Institute for Environmental Studies, Japan; Frontier Research Centre for Global Change, Japan), and of a regional climate model (RCM, PRECIS—Providing Regional Climates for Impact Studies —which had the Hadley Centre Climate Model (HadCM3) as the GCM), provided by the Indian Institute of Tropical Meteorology, India were used. These two models were selected because they were found to be relatively more suitable for simulating Indian climatic conditions (Rupa Kumar et al. 2006; Das et al. 2012). Since RCMs are generally more suitable than GCMs for simulating regional agricultural impacts, PRECIS outputs have been used in this study. PRECIS has been extensively used in climate change studies in India (INCCA 2010; NATCOM 2012). The greenhouse gas emission scenarios—A1b, A2 (more economic-driven with least concern for the environment, and with more GHG emissions), B1 and B2 (more consideration for environmental protection while deriving developmental policies, and with less GHG emissions)—for 2020, 2050 and 2080 were used for the analysis. This assessment is anticipated to provide inputs for deriving policies to help make rice cultivation more resilient to climate change.

2 Data and methods

2.1 Simulation analysis using InfoCrop

The impact of climate change was assessed using InfoCrop (Aggarwal et al. 2006), a crop growth process-based dynamic simulation model. It can simulate the effects of weather, soil, agronomic managements and major pests on crop growth and yield. The model considers different crop development and growth processes influencing the simulation of yield. The input varietal coefficients include thermal time and temperature thresholds for phenological stages (sowing to germination, germination to 50 % anthesis and grain filling period), potential grain weight, specific leaf area, maximum relative growth rate, and radiation-use efficiency. Crop management inputs in the model include dates of sowing, applications of nitrogen fertilizer, organic matter and irrigation; while soil input data include soil pH, slope, thickness of three soil layers, layer-wise texture, bulk density, saturated hydraulic conductivity, organic carbon, water holding capacity and permanent wilting point. Location-wise daily weather data (solar radiation, maximum and minimum temperatures, rainfall, wind speed and vapour pressure) are also used by the model to simulate crop performances.

The development of a crop is calculated by integrating the temperature-driven development rates from sowing to seedling emergence, and from seedling emergence to anthesis and storage organ filling phases. The dry matter production in this process is a function of radiation-use efficiency (RUE), photosynthetically active radiation, total leaf area index, and a crop/cultivar specific light interception coefficient. RUE is further governed by a crop-

specific response of photosynthesis to temperature, water, nitrogen availability and ambient CO_2. The net dry matter available each day for crop growth is partitioned as a function of the development stage. Temperatures influence the leaf area formation and senescence, and potential evapo-transpiration. A part of the storage organ becomes sterile if either maximum or minimum temperatures of the day deviate from their respective threshold values during the anthesis period and a few days thereafter. The rate of the storage organ filling is dependent upon temperature, potential filling rate and the level of available dry matter. The influence of rainfall is operated in the model through soil–water balance. Water stress—determined as the ratio of actual water uptake to potential transpiration—accelerates phenological development, decreases gross photosynthesis, alters the allocation pattern of assimilates to different organs, and accelerates the rate of senescence.

The InfoCrop model was calibrated and confirmed for different rice varieties grown in India (Mall and Aggarwal 2002). The present analysis was carried out for timely-sown irrigated as well as for rainfed rice during the monsoon season. While simulating baseline yields, regional variations in transplanting time, varietal variations and management were taken in to consideration. Rice transplanting generally follows the onset of monsoon across India, causing a one-month gap between the first transplanting in the first week of June in southern parts of India to the last transplanting in the third week of July in north-west India (Table 1). While the onset of the monsoon determined the transplanting time of the rainfed crop, the irrigated crop was transplanted at the optimal recommended time for a given region.

2.2 Processing of input data

2.2.1 Weather and soil data

The gridded daily data 1°×1° on rainfall, minimum and maximum temperatures (Indian Meteorological Department, IMD, Pune) for 22 years (1969–1990), and soil data (National Bureau of Soil Science and Land Use Planning, Nagpur, India, and Harmonized Global Soil Profile Dataset, Batjes 2008) on the above mentioned parameters were used. During the simulation process, solar radiation was calculated by the model, based on the Hargreaves method, which is reportedly best-suited for Indian conditions (Bandyopadhyay et al. 2008); and pedo-transfer functions were used to derive soil hydraulic characteristic coefficients.

2.2.2 Varietal coefficients

Dominant varieties (in short, medium and long duration groups) for a region were adopted for analysis assuming that farmers have optimized their variety and transplanting time (Table 1). Varietal coefficients on days to flowering and maturity, and grain size for dominant Indian varieties were taken from previous published studies (DRR 2009; DRD 2011).

2.2.3 Management

To represent farmers' field conditions, nitrogen was applied as urea @100 $kgNha^{-1}$ for irrigated crop, and @30 $kgNha^{-1}$ for rainfed rice at the time of transplanting and also 25 days after the transplanting, in equal spilt doses. Irrigated rice was provided irrigation whenever soil moisture level fell below 85 % of field capacity; while rainfed rice was transplanted when the soil moisture reached 85 % of field capacity at the onset of the

Table 1 State-wise statistics on total and irrigated rice area, major rice systems, transplanting windows, major varieties and duration of crop in different maturity groups and range of yield

State	Area – (Mega ha)	Area under irrigation (%)	Major rice systems	Transplantation times (Monsoon season)	Varieties	Days to maturity (range in each maturity group)	Yield Mg ha^{-1} (Range)
West Bengal	5.63	48.4	Irrigated, rainfed (lowland, upland), coastal saline, flood prone	June to mid- July,	Shatabdi, Laxmi, Kunti, Anjali, Annada, Heera	65–95; 105–115; 125–130	2.5–7.0
Punjab	2.80	99.5	Irrigated	June, July	PR-120, PS-1, PUSA1121, PR-113, PR-111	132, 158, 137, 113, 125	3.1–7.3
Uttar Pradesh	5.19	78.8	Upland, lowland flood prone, Irrigated	July	Narendra 97, Shusk Samrat, Pant Dhan 12, Sarjoo-52, Malaviya-36, Pusa Basmathi-1	90–105,120–130,130–135	3.5–6.5
Andhra Pradesh	3.44	96.8	Irrigated, rainfed low land, upland	June	Swarna, Samba Mashuri, MTU 9993, Nelluru Mashuri, Rasi	100–110; 120–125; 140–145	4.0–7.0
Orissa	4.37	46.8	Rainfed lowland, upland, irrigated, flooded, submerged	July	Anjali, Shatabdi, Ajay, Pooja	90–95; 110–115; 130–150	2.0–6.0
Tamil Nadu	1.85	93.3	Irrigated, Rainfed, upland	July	Improved White Ponni, ADT 36, Bhavani	105–120; 125–135; 135–140	2.5–4.5
Assam	2.50	5.3	Rainfed upland, Rainfed lowland	July, August	Kalyani, Heera, Annada, Krishna	62,68,111–115;140–145	2.0–5.6
Chattisgarh	3.67	32.7	Upland, lowland, irrigated; submergence prone	July second fortnight	Aditya, Vandana, Abhya, Vijeta	80–95, –, 120–130	2.5–6.0
Karnataka	1.49	74.7	Rainfed, irrigated, upland, submergence prone	June last week to July second fortnight	Jaya, MTU 1001, BR 2655, Rasi	110–115, 120–135, 140–145	4.5–6.0
Haryana	1.21	99.9	Irrigated	June, July	Ratna, CSR-10,HB-1,PB-1, PR-111	115–120, 130–140; 140–145	4.0–7.0
Bihar	3.21	57.2	Upland, lowland flood prone, Irrigated	July	Narendra 97, Shusk Samrat, Pant Dhan 12, Sarjoo-52,Malaviya-36, Pusa Basmathi-1	90–105,120–130; 135–140	3.0–6.5

Table 1 (continued)

State	Area – (Mega ha)	Area under irrigation (%)	Major rice systems	Transplantation times (Monsoon season)	Varieties	Days to maturity (range in each maturity group)	Yield Mg ha^{-1} (Range)
Maharashtra	1.47	26.4	Rainfed lowland, raifed upland, irrigated	June 2nd fortnight, July	Karjat 5, Karjat 6, Karjat 7, Shaydri-4, Ratnagiri-3	115–120, 125–130, 135–145	3.5–6.5
Jharkhand	1.00	2.2	Rainfed upland, Rainfed low land, Irrigated,	July	Anjali, Sadabahar, Hazaridhan,	90–95,105–110, 120–125	2.0–4.0
Gujarat	0.68	63.3	Irrigated, Rainfed, Rainfed upland	July	Jaya, Masuri, GR5, Gurjari, GR3	95–100, 115–120, 130–145	2.5–7.0
Madhya Pradesh	1.45	17.8	Rainfed upland, Rainfed low land, Irrigated,	July	IR 36, Pusa Sugandh 5, Swarna, Kranthi, MTU 1010	110–115, 120–125, 140–145	4.0–7.0
Kerala	0.23	72.2	Rainfed, water logged, irrigated	May last week, June	Uma, Jyothi, Kanchana	115–120, −, 155–160	2.5–4.0
All India	41.92	58.7 %	Irrigated and rainfed				

monsoon. Considering our limited understanding of the crop-pest interaction in future climates, it was assumed that crop was maintained free of pests and diseases in the field to quantify the direct impacts of climate change on crop yield.

2.3 Estimating impact of climate change

2.3.1 Estimating baseline yields

Yields of all the 22 years (1969–1990), simulated using the IMD gridded data (1×1°), were averaged grid-wise. District-wise yield was obtained as a sum of the weighted yield from each grid fraction in the district. The simulated yield output was then calibrated to district yield (district-wise mean yield of the 2000–2007 period; DES 2012) to get the baseline yields.

2.3.2 Simulating yields in future scenarios

The temperature (minimum and maximum) and rainfall data from MIROC3.2.HI—A1b, B1 for 2020, 2050 and 2080 scenarios; and the PRECIS —A1b, A2 and B2 for 2020 and 2080 scenarios—were used for the analysis. The spatial resolution of MIROC3.2.HI was 1.125°× 1.125° and that of PRECIS was 0.44°×0.44°. Since the observed weather was in 1°×1° resolution, the GCM and RCM outputs were rescaled to 1°×1° resolution for easy comparison and aggregation of the crop model outputs. To get the future climate, the differences between the model-derived baseline and the respective scenario temperatures (minimum and maximum in °C) and rainfall (in % change) were added to the observed gridded temperature and rainfall data. Projected CO_2 level for respective scenario —2020-410 (B1), 418 (A1b); 2050-482 (B1), 522 (A1b); 2080- 530 (B1), 552 (B2), 639 (A1b) and 682 (A2)—was also included in the model for simulations. The district-wise yields in future climates were calculated as per the method used to get the baseline yields. The impact of climate change on the mean yield of 22 years is expressed as the percentage change from baseline yields.

2.4 Simulating adaptation gains in future scenarios

Several low cost and easy-to-adopt options were tested independently, or in combination, as adaptation options to assess adaptation gains in the rice yields in future climates. Strategies included genotypic and agronomic management options such as i) use of short, medium and long duration varieties; ii) change in sowing time which included early (up to 15 days) as well as late sowing (up to 15 days), relative to the current optimal sowing time; iii) rescheduling irrigation and fertilizer, and iv) rescheduling irrigation and fertilizer with 25 % additional nitrogen to suit rice phenology in future climates. The combination which gave the highest yield in each grid was taken as the best suitable adaptation option. The net change in mean yield for all the 22 years was expressed as the relative difference with the mean baseline yield. The net vulnerability in the respective scenarios was derived from the following:

$$\text{Vulnerability} = \text{Impact}(\text{yield loss due to climate change}) - \text{Adaptation gain}$$

In instances where impact on yield was positive, simulations were run for similar adaptation strategies for quantifying additional benefits to represent net positive impacts, maximized with additional adaptation measures. In all, about 4.37 million simulations (22 years*(10 scenarios + baseline)* 355 grids*irrigated and rainfed crop* 5 varieties* 4

transplanting dates; + 1 million simulations for rescheduling of nitrogen and irrigation and for 25 % additional nitrogen) were carried out for the entire analysis.

3 Results and discussion

3.1 Impact of climate change on rice productivity

3.1.1 Impact on irrigated monsoon season rice

Irrigated rice yields have been projected to reduce progressively in 2020, 2050 and 2080 climate scenarios, with spatial variation in magnitude of impacts (Fig. 1). On an all-India basis, climate change is projected to reduce irrigated rice yields by ~4 % in 2020 (2010–2039), ~7 % in 2050 (2040–2069) and ~10 % in 2080 (2070–2099) scenarios. At regional level, states like Punjab, Haryana and Rajasthan are projected to lose more yields (6–8 %) in the 2020 scenario. Similarly in the 2050 scenario, projected yield loss is expected to stand at 15–17 % in the above three states, while in some parts of central India (Maharashtra and Madhya Pradesh) the yield loss may be >5 %. However, towards the end of the century, many regions in India are projected to lose yields in the range of 6–21 %. In general, the range of impacts due to different climate scenarios is anticipated to be narrower in the 2020s, but is likely to progressively increase in later climate scenarios. For instance, in northwest India, yield loss may be 5–12 % in 2020, 8–22 % in 2050 and 23–30 % in the 2080 scenarios (Fig. 2).

While variation in atmospheric CO_2 concentrations in different emission scenarios cause different positive impacts on rice yields; those of temperatures and rainfall change may benefit or reduce yields depending on current values. Under irrigated conditions, the uncertainty was mainly due to the magnitude of change in temperature rather than rainfall. Though uncertainties in crop productivity can be due to various reasons, the response of crop to temperature is reported to be important (Lobell and Burke 2008). More reduction in irrigated rice yields have been projected for areas with current seasonal (June-September) mean minimum temperatures of >23 °C, as in parts of central (Madhya Pradesh, eastern Rajasthan and Maharashtra), north (Uttar Pradesh, Bihar and Jharkhand) and northeast regions of India. These temperatures are projected to increase by ~1.4–2 °C in 2020 and

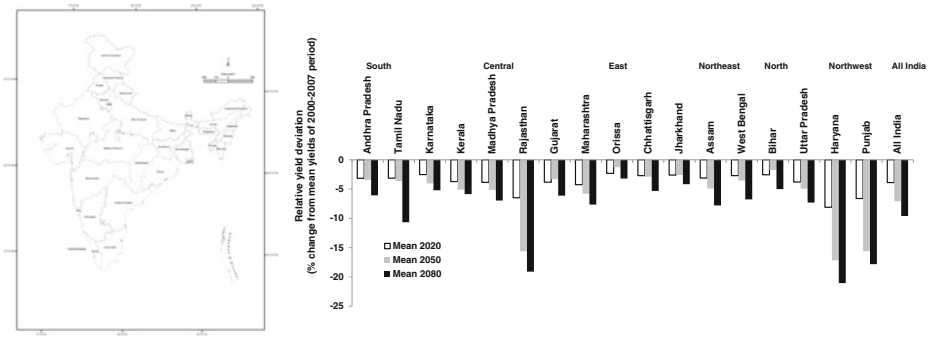


Fig. 1 Impact of climate change on irrigated rice yields in different regions (covering several states) in India. The mean impacts are for three time frames, viz. 2020 (2010–2039), 2050 (2040–2069), and 2080 (2070–2099). Major states falling in each region are labeled. *Left panel* provides the state boundaries and names for reference

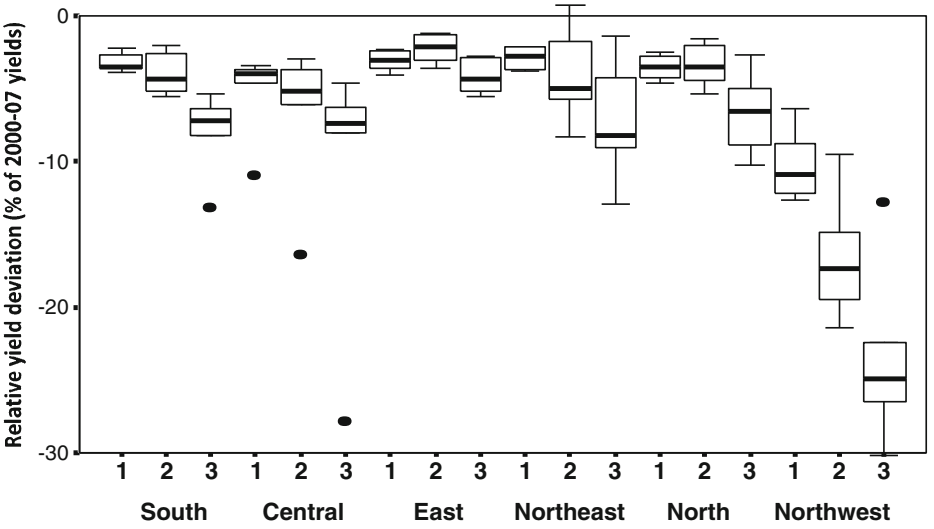


Fig. 2 Range of climate change impact on irrigated rice yields in different regions of India. *Box-whisker plots* indicate the extent of uncertainty among different scenarios (A1b, A2, B1 and B2 scenarios of MIRO 3.2HI and PRECIS) for 2020 (1), 2050 (2) and 2080 (3) periods in different regions of India. The *dots* represent extreme values for impacts

~3–3.5 °C in the 2080 scenario. Seasonal mean maximum temperatures in these areas range from 31 to 34 °C; and a projected increase of ~1–1.7 °C in 2020 and ~3–3.5 °C in the 2080 scenarios may cause heat stress on crop. Projected rise in temperatures for north western India are even higher (1.8–2.8 °C in 2020 and 4–5.3 °C in 2080 scenarios); and therefore, future temperatures constrain higher productivity of irrigated rice in these areas. However, regions with currently low temperatures may benefit due to a rise in temperature as thermal regimes shift towards an optimal range. Rice, a tropical crop, is well adapted to high temperature conditions, with temperature optima of 36 °C for vegetative development, 33 °C for reproductive development, and 23–26 °C for maximum grain yield (Baker et al. 1995). Temperatures above 33 °C coinciding with anthesis (Prasad et al. 2006) can affect anther dehiscence, shedding and germination of pollen affecting pollination, apart from reduced grain size, grain filling duration and therefore grain yield. Even though *japonica* and *indica* rice varieties did not differ on their upper threshold of temperature (Snyder 2000; Prasad et al. 2006), *indica* types are reported to be more sensitive to night temperatures >19 °C (Snyder 2000). On the whole, increase in minimum temperatures are detrimental to rice yields (Cheng et al. 2010) mainly due to reduced translocation of C and N, thus offsetting positive impacts of elevated CO_2 on rice crop.

3.1.2 Impact on rainfed monsoon season rice

The mean impact of emission scenarios indicate that rainfed rice productivity in India may reduce by ~6 % in the 2020 scenario (Fig. 3). However, in the 2050 and 2080 scenarios these yields are projected to decrease marginally (<2.5 %). Climate change is expected to benefit rainfed rice in Tamil Nadu and Andhra Pradesh by 3–22 %; while in Maharashtra, Orissa, Chhattisgarh and Assam, yields are expected to reduce by 8–10 % in the 2020 scenario. In the 2050 scenario, rainfed rice yields in parts of south India (Andhra Pradesh, Tamil Nadu and Karnataka) are projected to benefit by ~10–15 %. However, in all other regions,

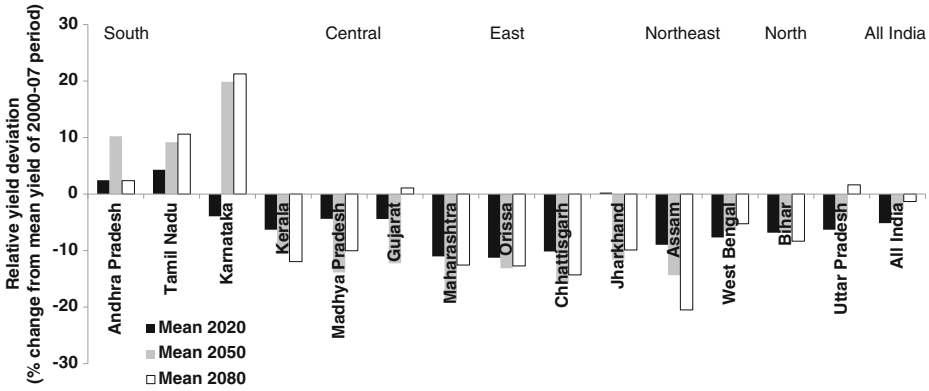


Fig. 3 Impact of climate change on rainfed rice yields in different regions (covering several states) of India. The mean impacts are for three time-frames, viz. 2020 (2010–2039), 2050 (2040–2069), and 2080 (2070–2099). Major states falling in each region are labeled

projected reduction stands at 6–18 % with more impacts on central and east India. Almost similar trends are projected for the 2080 scenario too.

The range of impacts is wider in currently high rainfall areas—such as the northeast region (Fig. 4)—and in areas projected to witness significant changes in rainfall in future climates—such as the southern region. The range of impacts is expected to be wider in rainfed over that of irrigated conditions, mainly because of uncertainties inherited from the climate model projections on rainfall changes. The sources of uncertainty include the emission scenario and climate model's ability to capture regional climatic patterns—particularly that of the Indian monsoon (Rajeevan and Nanjundiah 2010; Kodra et al. 2012).

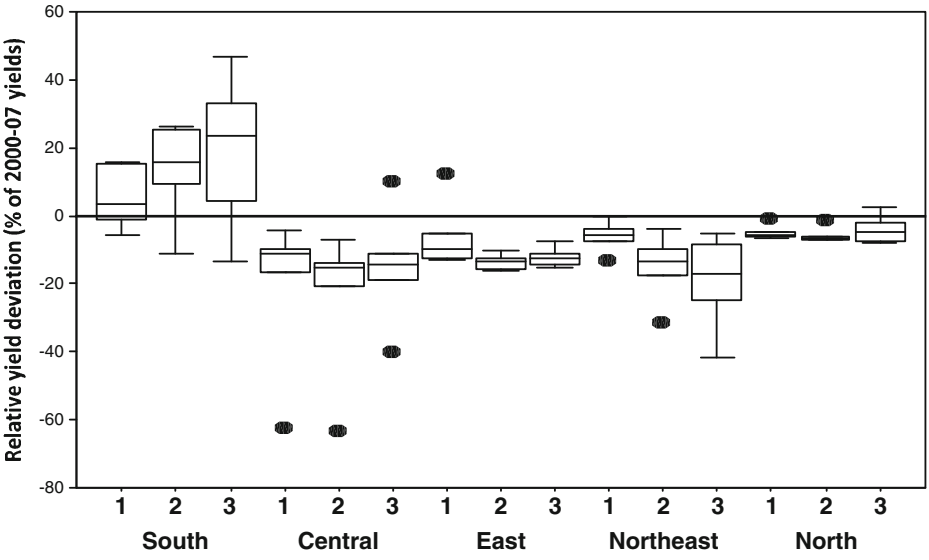


Fig. 4 Range of climate change impact on rainfed rice yields in different regions of India. *Box-whisker plots* indicate the extent of uncertainty among different scenarios (A1b, A2, B1 and B2 scenarios of MIRO 3.2HI and PRECIS) for 2020 (1), 2050 (2) and 2080 (3) periods in different regions of India. The *dots* represent extreme values for impacts

In addition to CO_2 and temperature, change in rainfall amount, intensity and distribution will also have significant influence in India since ~42 % area under rice is rainfed. An increase in rainfall in scarce rainfall zones can improve rainfed rice yields and offset impacts of rise in temperature to a certain extent, as reported for sorghum (Srivastava et al. 2010). Rain fed rice yield is projected to increase in low rainfall (400–800 mm of monsoon rainfall) regions of south (Tamil Nadu, southern Andhra Pradesh) and central (central Maharashtra and parts of Madhya Pradesh) India, where rainfall is projected to increase by 5–15 % in the 2020 scenario, and by 15–25 % in the 2080 scenario. Even a 10 % reduction in rainfall can reduce the rainfed rice yields in currently low-rainfall areas of central (western Madhya Pradesh, eastern Rajasthan) and north (parts of western Uttar Pradesh) India. On the other hand, high rainfall (>1,000 mm of monsoon rainfall) areas in central (Madhya Pradesh), east (Orissa, Jharkhand and Chhattisgarh) and northeast regions of India may lose yields due to a projected increase in rainfall (5–15 % by 2020 and 10–35 % by 2080).

3.2 Adaptation to climate change

3.2.1 Irrigated monsoon season rice

Sowing rice varieties with a restored crop duration in future climates (improved varieties), similar to that of rice varieties in the current climate, can offset the negative impacts of climate change in the 2020 and 2050 scenarios (Fig. 5a). Yet, irrigated rice in the 2080 scenario may still be vulnerable by ~3 % of current yields. Current varieties may perform better under increased input and efficiency regimes, offsetting yield loss in the 2020 scenario. But in later periods, viz. 2050 and 2080 scenarios, irrigated rice yield may be vulnerable by −6 and −8 %, respectively. Growing improved varieties with efficient application and re-scheduling of inputs suiting crop phenology, however, can reduce the yield loss. In addition, using a higher dose of fertilizers (about 25 % higher than that currently applied by farmers) can increase all-India irrigated rice yields by about 17 % over current values in the 2020 scenario, and by 14 % and 8 % in the 2050 and 2080 scenarios, respectively. These results indicate that the impact of climate change on irrigated rice can be offset with either agronomical management or with improved varieties in shorter periods (the 2020 scenario); but in the longer periods (2050 and 2080 scenarios), growing improved varieties, coupled with efficient crop management, will become essential to enhance productivity for meeting increased demand.

3.2.2 Rainfed monsoon season rice

Adaptation to climate change by sowing improved rice varieties can not only reduce negative impacts but also provide improved productivity—ranging from 1 % to 5 % in climate change scenarios (Fig. 5b). Even with current varieties, increasing input use efficiency and providing additional nitrogen fertilizers can improve yields by ~15 % in the 2020, and by ~25 % in the 2050 and later scenarios. In addition, adopting improved varieties can further improve yields by ~20 % in the 2020, and by ~35–38 % in the 2050 and later scenarios. Adaptation gains are projected to be higher in rainfed than in irrigated conditions, due to current low yields from rainfed rice. Growing current or improved varieties in low-input management conditions will not be beneficial as current varieties in rainfed areas can give up to 40 % improved yields, if recommend fertilizers are used (Fig. 5b) along with better management of available water.

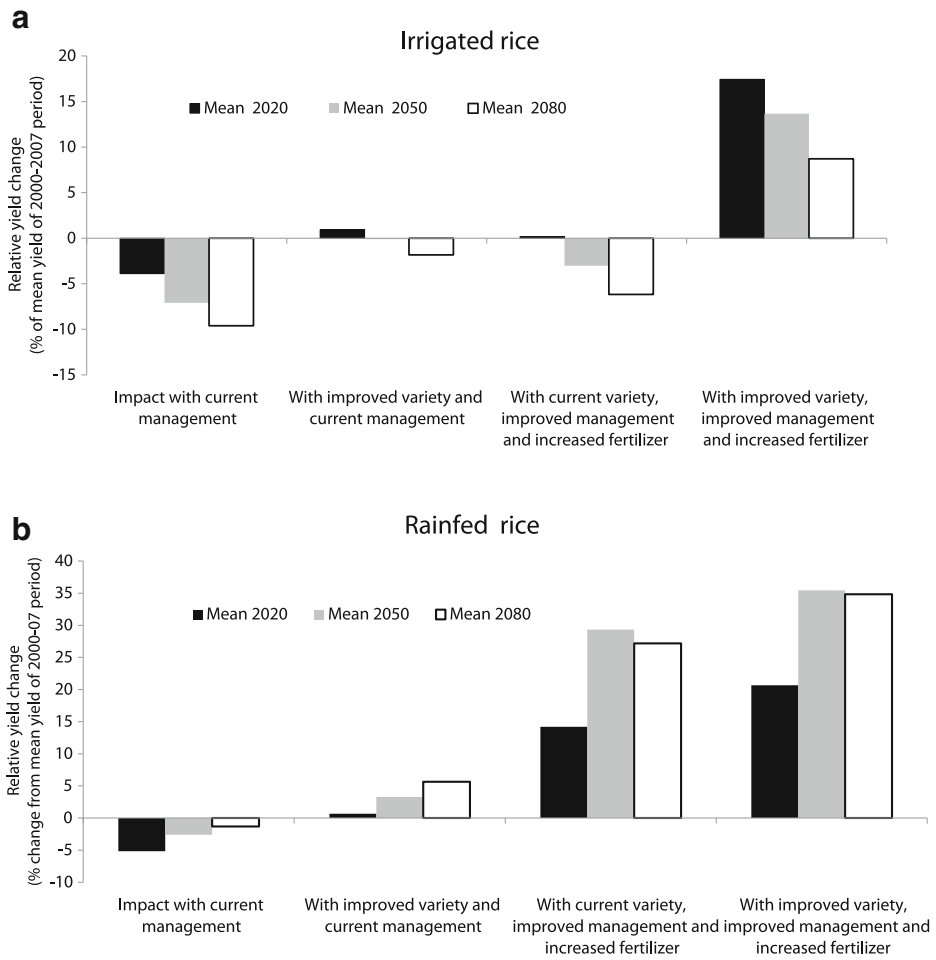


Fig. 5 Impacts, with and without adaptation, on (**a**) irrigated and (**b**) rainfed rice yields in India in 2020 (2010–2039), 2050 (2040–2069) and 2080 (2070–2099) climate scenarios. Each *bar* is a mean of the data from simulations for different climate scenarios for respective time period; and has been generated from the mean results for 22 years of the baseline period (1969–1990) and respective scenario time. The relative deviations are worked out for each grid and up-scaled to the national level. The net adaptation gain below zero indicates the vulnerability

3.3 Vulnerability of rice crop to climate change

The magnitude of adaptation gains and therefore that of net vulnerability vary with climate scenario and also with adaptation option (Figs. 6 and 7). Climate change is likely to cause three basic types of impacts—viz., i) regions that are adversely affected can have net improvement in yield in future scenarios with adaptation (most of irrigated rice areas and rainfed rice areas of south-east, parts of west and north India fall in this category); ii) regions that are adversely affected may still remain vulnerable despite adaptation gains (rainfed rice areas in parts of south-west, central and northeast India fall in this category); and iii) rainfed regions (with currently low rainfall) likely to benefit with increase in rainfall and adaptation can enhance this benefit (these areas include parts of south-west and central India). Regions

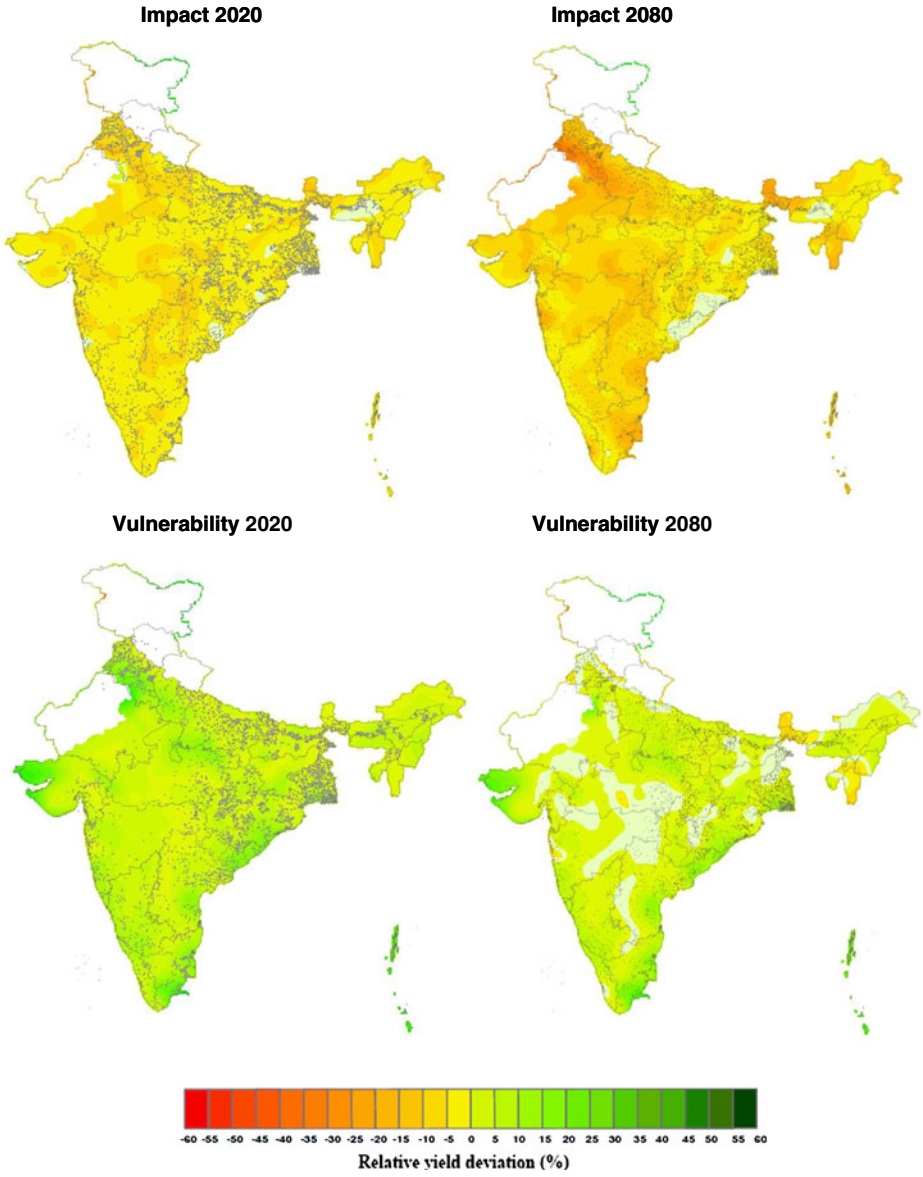


Fig. 6 Spatial variation in climate change impact on irrigated rice yields and net vulnerability in India. The maps are generated from the mean results for the 22 years of the baseline period (1969–1990); and the scenarios of 2020 (2010–2039) and 2080 (2070–2099). The relative deviations are worked out for each grid for generating the surface map. In case of vulnerability, positive values indicate a gain in yield with adaptation. The *dots* indicate rice growing areas in India, with each dot representing 10,000 ha. *Foot note*: Mean seasonal minimum temperatures are ~23–24 °C or more in parts of central (Madhya Pradesh, eastern Rajasthan and Maharashtra), north (Uttar Pradesh, Bihar and Jharkhand) and northeast regions of India, which are projected to increase by ~1.4–2 °C in 2020 and ~3–3.5 °C in the 2080 scenarios. Seasonal mean maximum temperatures in these areas range from 31 to 34 °C; and are projected to increase by ~1–1.7 °C in the 2020 and by ~3–3.5 °C in the 2080 scenario. In north western India, the projected rise in temperatures are ~1.8–2.8 °C in the 2020 and 4–5.3 °C in the 2080 scenarios

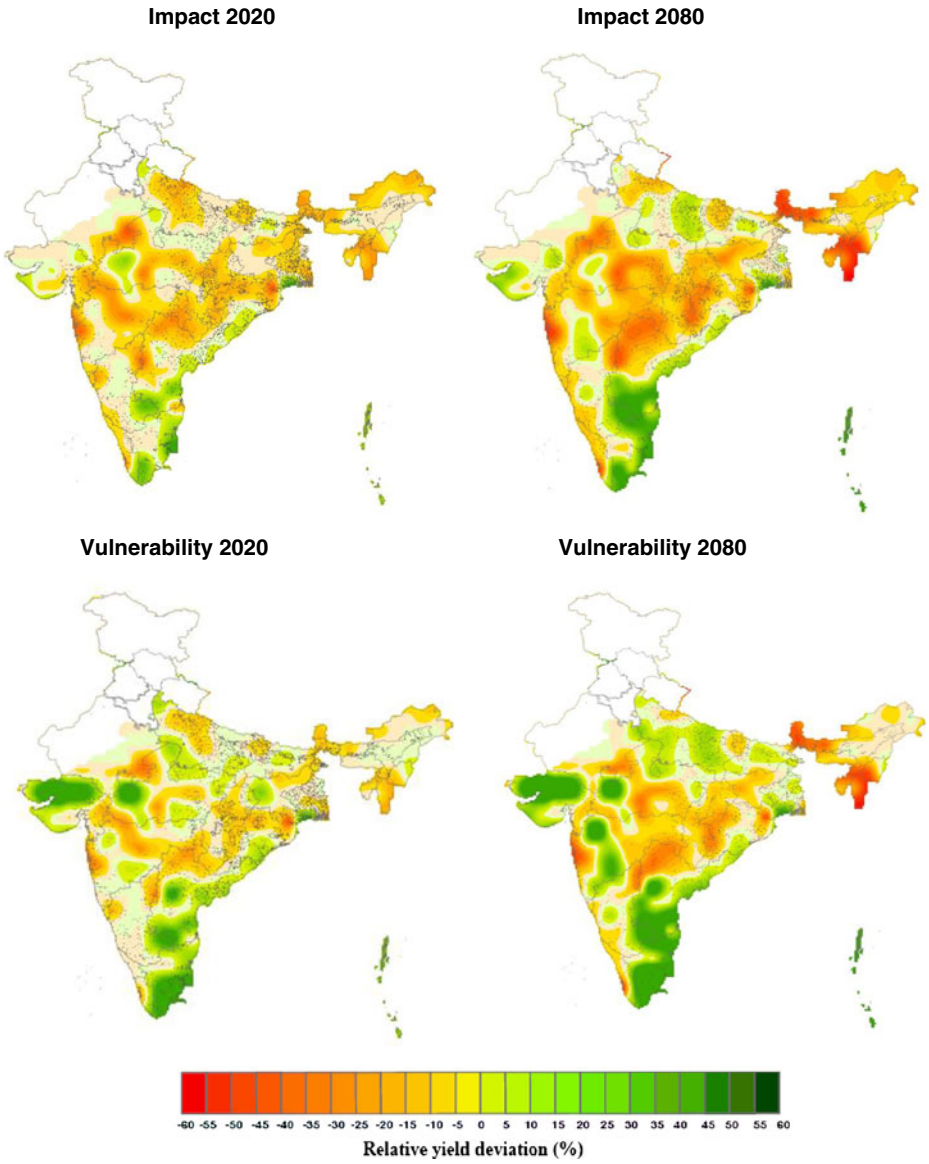


Fig. 7 Spatial variation in climate change impacts on rainfed rice yields and net vulnerability in India. The maps are generated from the mean results for the 22 years of the baseline period (1969–1990); and the scenarios of 2020 (2010–2039) and 2080 (2070–2099). The relative deviations are worked out for each grid for generating the surface map. In case of vulnerability, positive values indicate a gain in yield with adaptation. The dots indicate rice growing areas in India with each dot representing 10,000 ha. *Foot note*: Monsoon rainfall is projected to increase in parts of south (Tamil Nadu, southern Andhra Pradesh; 500–1,000 mm current rainfall), central (central Maharashtra and parts of Madhya Pradesh; 500–1,000 mm current rainfall), east (Orissa, Jharkhand and Chhattisgarh; 1,000–1,500 mm current rainfall) and northeast (1,500–3,500 mm current rainfall) India by 5–15 % in the 2020 and by 10–35 % in the 2080 scenarios; while it is projected to reduce by 10 % in parts of central (western Madhya Pradesh, eastern Rajasthan; current rainfall 500–800 mm) and north (parts of western Uttar Pradesh; current rainfall 500–800 mm) India

falling in the vulnerable category despite suggested adaptation options require more intensive, specific and innovative interventions beyond crop management and crop improvement strategies tested in this study. A better understanding of genetic variability for grain filling duration and multiple-stress tolerance can reduce negative impacts. Genotypic variations for responses to CO_2 and temperature (Moya et al. 1998), and cultivars with heat tolerance in terms of production, shed and viability of pollen, and filled grains are available (Prasad et al. 2006). Cultivars shedding pollen in the early hours of the day can avoid extreme temperature events coinciding with flowering. In addition, efforts to convert rice photosynthesis from C3 to C4 mechanism (Sheehy et al. 2007) can improve rice yields even at current CO_2 levels.

The InfoCrop model did not significantly respond to an increase in CO_2 levels in resource poor environments, such as rainfed low fertility situations, since genotypes with higher yield potential fail to express them in low fertility levels (Aggarwal et al. 1997). Even though nitrogen use efficiency increases at elevated CO_2 (Leakey et al. 2009), in low fertility situations nitrogen may become limiting and higher doses of it, particularly in later phases of growth, may be needed (Weerakoon et al. 2005) to reap the CO_2 fertilization benefits as well as to maintain the C/N ratio and the grain nitrogen concentration (Kim et al. 2003b; Lieffering et al. 2004). In addition, losses due to volatilization and leaching also need be reduced since climate change causes increase in temperature and more intense rainfall events. In the present study, the use of an additional 25 % of nitrogen led to higher harvests.

The major uncertainties of this assessment include non-consideration of future socio-economic trends—such as land use; technological improvements, and changes in water, soil fertility, pests and disease scenarios. In spite of these, this analysis provides vital information about the impacts, adaptation gains and vulnerable areas for enabling better preparedness to reduce the negative impacts of climate change.

4 Summary and conclusions

- The study projects a progressive reduction in irrigated rice yields due to climate change towards the end of the century, if no adaptation is followed; while for rainfed rice, negative impacts likely to reduce with time due to projected increase in rainfall in many areas. However, spatio-temporal variations exist for the magnitude of impacts.
- Adaptation to climate change can not only minimize negative impacts, but can also substantially increase productivity. As a result, current varieties may be able to perform better under increased input and efficiency regimes, offsetting yield loss in near future; but in the longer run, developing suitable varieties coupled with improved and efficient crop husbandry will become essential. For irrigated rice, genotypic and agronomic improvements will become crucial; while for rainfed conditions, improved management and additional fertilizers will be needed in near future.
- Three basic types of climate change influence on rice productivity will be i) regions that are adversely affected can have net gain in productivity with adaptation; ii) regions that are adversely affected will remain vulnerable despite adaptation strategies considered in this study; and iii) rainfed regions (with currently low rainfall) likely to benefit due to increase in rainfall can further benefit by adaptation. Regions falling in the vulnerable category even after suggested adaptation to climate change will require more intensive, specific and innovative adaptation options.

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References

Aggarwal PK, Mall RK (2002) Climate change and rice yields in diverse agro-environments of India. II. Effect of uncertainties in scenarios and crop models on impact assessment. Clim Chang 52:331–343

Aggarwal PK, Kropff MJ, Cassman KG, ten Berge HFM (1997) Simulating genotypic strategies for increasing rice yield potential in irrigated, tropical environments. Field Crop Res 51:5–17

Aggarwal PK, Kalra N, Chander S, Pathak H (2006) InfoCrop: a dynamic simulation model for the assessment of crop yields losses due to pests and environmental impact of agro-ecosystems in tropical environments I Model description. Agric Syst 89:1–25

Baker JT (2005) Yield responses of southern US rice cultivars to CO_2 and temperature. Agric For Meteorol 122:129–137

Baker JT, Boote KJ, Allen LH Jr (1995) Potential climate change effects on rice: carbon dioxide and temperature. In: Rosenzweig C et al (eds) Climate change and agriculture: analysis of potential international impacts. ASA Spec. Publ.59. ASA, CSSA and SSSA, Madison, pp 31–47

Bandyopadhyay A, Bhadra A, Raghuwanshi NS, Singh R (2008) Estimation of monthly solar radiation from measured air temperature extremes. Agric For Meteorol 148:1707–1718

Batjes NH (2008) ISRIC-WISE harmonized global soil profile dataset (V. 3.1). Report 2008/2, ISRIC-World Soil Information. Wageningen. The Netherlands

Cheng W, Sakai H, Yagi K, Hasegawa T (2010) Combined effects of elevated CO_2 and high night temperature on carbon assimilation, nitrogen absorption, and the allocation of C and N by rice (*Oryza sativa* L.). Agric For Meteorol 150:1174–1181

Das L, Annan JD, Hargreaves JC, Emori S (2012) Improvements over three generations of climate model simulations for eastern India. Clim Res 51:201–216

DES (2012) Department of economics and statistics, agricultural statistics at a glance 2011. Ministry of Agriculture, Government of India

DRD (2011) Notified rice varieties in India during 1996–2010. Directorate of Rice Development. Ministry of Agriculture, Government of India. p 41

DRR (2009) Annual reports (2001-09). All India Coordinated Rice Improvement Programme (ICAR), Directorate of Rice Research, Hyderabad, India

Fan S, Chan KC (2005) Is small beautiful? Farm size, productivity, and poverty in Asian agriculture. Agric Econ 32:135–146

FAO (2009) Global agriculture towards 2050. High level expert forum-how to feed world in 2050. Rome, Italy. http://www.fao.org/fileadmin/templates/wsfs/docs/Issues_papers/HLEF2050_Global_Agriculture.pdf

FAOSTAT (2012) http://faostat3.fao.org/home/index.html#VISUALIZE_BY_DOMAIN

Hatfield JL, Boote K, Kimball BA, Ziska LH, Izaurralde RC, Ort D, Thomson AM, Wolfe D (2011) Climate impacts on agriculture: implications for crop production. Agron J 103:351–370

INCCA (2010) Climate change and India: A 4×4 assessment. Ministry of Environment and Forests, Government of India, New Delhi, India. http://envfor.nic.in/downloads/public-information

IPCC (2007) IPCC fourth assessment report: climate change 2007(AR4). IPCC, Geneva

IRRI (2012). International Rice Research Institute, www.irri.org

Kim HY, Lieffering M, Kobayashi K, Okada M, Miura S (2003a) Seasonal changes in the effects of elevated CO_2 on rice at three levels of nitrogen supply: a free air CO_2 enrichment (FACE) experiment. Glob Chang Biol 9:826–837

Kim HY, Lieffering M, Kobayashi K, Okada M, Mitchell MW, Gumpertz M (2003b) Effects of free-air CO_2 enrichment and nitrogen supply on yield of temperate paddy rice crops. Field Crops Res 83:261–270

Kodra E, Ghosh S, Ganguly AR (2012) Evaluation of global climate models for Indian monsoon climatology. Environ Res Lett 7(014012):7. doi:10.1088/1748-9326/7/1/014012

Krishnan P, Swain DK, Baskar C, Nayak SK, Dash RN (2007) Impact of elevated CO_2 and temperature on rice yield and methods of adaptation as evaluated by crop simulation studies. Agric Ecosyst Environ 122:233–242

Leakey ADB, Ainsworth EA, Bernacchi CJ, Rogers A, Long SP, Ort DR (2009) Elevated CO_2 effects on plant carbon, nitrogen, and water relations: six important lessons from FACE. J Exp Bot 60:2859–2876

Lieffering M, Kim HY, Kobayashi K, Okada M (2004) The impact of elevated CO_2 on the elemental concentrations of field grown rice grains. Field Crops Res 88:279–286

Lobell DB, Burke MB (2008) Why are agricultural impacts of climate change so uncertain? The importance of temperature relative to precipitation. Environ Res Lett 3:1–8

Mall RK, Aggarwal PK (2002) Climate change and rice yields in diverse agro-environments of India. I. Evaluation of impact assessment models. Clim Chang 52:315–330

Matsui T, Omasa K, Horie T (2000) High temperature at flowering inhibits swelling of pollen grains, a driving force for anther dehiscence in rice (*Oryza sativa* L.). Plant Prod Sci 3:430–434

Matthews RB, Wassmann R (2003) Modelling impacts of climate change and methane emission reductions on rice production: a review. Eur J Agron 19:573–598

Moya TB, Ziska LH, Namuco OS, Olszyk D (1998) Growth dynamics and genotypic variation in tropical, field-grown paddy rice (*Oryza sativa* L.) in response to increasing carbon dioxide and temperature. Glob Chang Biol 4:645–656

NATCOM (2012) India's Second National Communication to UNFCCC, Ministry of Environment and Forests, Government of India, New Delhi, India. http://envfor.nic.in/downloads/public-information/India%20Second%20National%20Communication%20to%20UNFCCC.pdf

Paroda RS, Kumar P (2000) Food production and demand in South Asia. Agril Econ Res Rev 13:1–24

Prasad PVV, Boote KJ, Allen LH Jr, Sheehy JE, Thomas JMG (2006) Species, ecotype and cultivar differences in spikelet fertility and harvest index of rice in response to high temperature stress. Field Crops Res 95:398–411

Rajeevan M, Nanjundiah R (2010) Coupled model simulations of twentieth century climate of the Indian summer monsoon. In: Current trends in science, Indian Academy of Sciences Publication, Bangalore, pp 537–567 (available at http://www.ias.ac.in)

Rupa Kumar K, Sahai AK, Krishna Kumar K, Patwardhan SK, Mishra PK, Revadekar JV, Kamala K, Pant GB (2006) High-resolution climate change scenarios for India for the 21st century. Curr Sci 90:335–345

Sheehy JE, Mitchell PL, Hardy B (2007) Changing new pathways to C4 rice. IRRI, Los Banos

Smit B, Skinner MW (2002) Adaptation options in agriculture to climate change: a typology. Mitig Adapt Strateg Glob Chang 7:85–114

Snyder AM (2000) The effects of elevated carbon dioxide and temperature on two cultivars of rice. M.S. thesis. Univ. of Florida, Gainsaville

Srivastava A, Naresh Kumar S, Aggarwal PK (2010) Assessment on vulnerability of sorghum to climate change in India. Agric Ecosyst Environ 138:160–169

Weerakoon WMW, Ingram KT, Moss DN (2005) Atmospheric CO_2 concentration effects on N partitioning and fertilizer N recovery in field grown rice (*Oryza sativa* L.). Agric Ecosyst Environ 108:342–349